

Ideation Phase

Empathize & Discover

Date	31 January 2025
Team ID	LTVIP2026TMIDS64949
Project Name	Online Payments Fraud Detection using Machine Learning
Maximum Marks	4 Marks

Empathy Map Canvas:

Empathy Map Canvas: An empathy map helps the team deeply understand users by capturing what they think, feel, see, hear, say, and do. It enables building solutions grounded in real user needs and constraints..

● THINK & FEEL

- Is my online payment secure?
- What if someone misuses my card details?
- I hope the system detects fraud quickly.
- I don't want to lose my money.
- Online transactions should be safe and fast.
- I trust digital payments but worry about fraud.

● HEAR

- "There are many online payment scams."
- "Use secure and trusted payment apps."
- "Enable OTP verification."
- "Banks monitor suspicious transactions."
- News about increasing cyber fraud cases.

● SEE

- OTP verification screens.
- Payment confirmation messages.
- Fraud alert notifications from bank.
- Transaction history in banking app.

- Security badges and encryption symbols.
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● SAY & DO

- Checks transaction details carefully.
 - Verifies OTP before confirming payment.
 - Contacts bank if suspicious activity occurs.
 - Uses strong passwords.
 - Enables two-factor authentication.
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● PAIN

- Fear of money loss.
 - Delay in refund process.
 - Unauthorized transactions.
 - Complex complaint procedures.
 - False fraud alerts blocking genuine payments.
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● GAIN

- Secure and safe transactions.
- Instant fraud detection alerts.
- Quick refund process.
- Trust in digital payment systems.
- Smooth and uninterrupted online payments.

How Our Solution Helps: The Machine Learning–based Online Payment Fraud Detection System analyzes transaction behavior patterns, detects anomalies in real-time, and flags suspicious activities instantly. This enhances transaction security, reduces financial loss, and builds trust for customers, banks, and payment platforms.

